WEEK 3 – TUTORIAL

Objectives:

The aim of this tutorial is to equip students with practical skills in identifying, analysing, and engaging stakeholders, as well as effectively gathering project requirements in an IT project context. Students will learn to recognise the importance of stakeholder influence and interest, apply stakeholder analysis tools, and connect stakeholder needs to project deliverables. The session will also explore and discuss key requirement gathering strategies (such as interviews, workshops, observation, etc.) and stakeholder analysis strategies (including Power–Interest Grids), ensuring students can select and apply the most appropriate techniques for different project scenarios.

Tutorial Activities

- Requirement Gathering and Stakeholder Analysis
- Lecture Recap

Learning outcomes

By the end of this tutorial, students will be able to:

- LO1. Understand each of the core project management knowledge areas
- LO4. Develop basic proficiency in project management tools and practices

Part A – Project Methodology

Your organization has been tasked with developing a new mobile application for a healthcare provider. The project involves multiple stakeholders including doctors, patients, and regulatory bodies, and must comply with strict legal and security standards.

- Compare Agile and Waterfall methodologies in the context of this project.

Aspect	***Agile***	***Waterfall***
Approach	*Iterative and incremental; delivers working software in short sprints*	*Sequential and linear; all requirements gathered upfront*
Flexibility	*Highly flexible. Changes can be incorporated during the project*	*Low flexibility. Changes after the design phase are costly*
Stakeholder Involvement	*Continuous stakeholder feedback and collaboration*	*Stakeholders mainly involved at the start (requirements) and end (delivery)*
Documentation	*Lightweight and adaptive; documentation evolves with the project*	*Heavy upfront documentation before moving to the next phase*
Risk Management	*Risks identified and mitigated iteratively during each sprint*	*Risks identified early, but may be overlooked if they emerge mid-project*
Compliance & Quality	*Can integrate compliance reviews into each sprint*	*Compliance checked mostly at final testing/validation stage*

- Which methodology (or hybrid approach) would you recommend and why?

A Hybrid Agile-Waterfall (also called "Water-scrum-fall") approach.

- *Reason:*
  - *Agile allows incremental delivery, user feedback, and adaptability for changing healthcare needs.*
  - *Waterfall ensures structured compliance, documentation, and traceability, which are critical for healthcare regulations (HIPAA, GDPR, etc.).*
- *Implementation Example:*
  - *Use Waterfall for initial requirements gathering, compliance design, and regulatory sign-off.*
  - *Switch to Agile for iterative development and user feedback cycles.*

- Identify at least three risks associated with your chosen methodology and suggest mitigation strategies.

Risk	***Impact***	***Mitigation Strategy***
Regulatory non-compliance 违反法规	*Legal penalties, project delays*	*Involve compliance officers from day one, conduct reviews at the end of every sprint*
Stakeholder fatigue in Agile	*Reduced engagement and unclear requirements*	*Rotate review sessions, prioritize high-impact stakeholders, keep sprint reviews concise*
Integration challenges between Agile iterations and Waterfall documentation	*Misalignment of technical and compliance deliverables*	*Use a dedicated integration manager to align Agile outputs with compliance documents*

Part B – Requirement Gathering and Stakeholder Analysis

You are working for a company and it is developing an **Online Course Registration System** for a university. In small groups,

- List at least 6 stakeholders
  - *Registrar - Oversees enrolment process, compliance with academic policies.*
  - *Students - End-users who will register/drop courses.*
  - *Lecturers - Need updated class lists, schedules, and enrolment data.*
  - *IT Manager - Responsible for technical infrastructure, integration, and security.*
  - *Finance Officer - Handles tuition fee processing linked to enrolment.*
  - *University Administration - Executive body approving project direction and funding.*
- Place them in a Power/Interest Matrix (you will have to research about this in the first few minutes and discuss amongst your group)

	High Power	***Low Power***
High Interest	*Registrar – Closely Manage*	*Students – Keep Informed*
	IT Manager – Closely Manage	*Lecturers – Keep Informed*
Low Interest	*University Administration – Keep Satisfied*	*Finance Officer – Monitor*

- Assign one of your group member to be a Project Manager, while the others assume the other stakeholder positions. Your task is to now:
  - PM is to gather requirements about the project from the various stakeholders.
    - *Registrar*
      - *Needs: Ability to approve course offerings, enforce prerequisites.*
      - *Constraints: Must align with university regulations.*
      - *Expectations: System should be accurate and reduce manual processing.*
    - *Students*
      - *Needs: Search, register, drop courses online.*
      - *Constraints: Access only during open enrolment periods.*
      - *Expectations: Fast, mobile-friendly, and error-free.*
    - *Lecturers*
      - *Needs: View class rosters, send announcements.*
      - *Constraints: Limited editing rights to avoid schedule conflicts.*
      - *Expectations: Integration with learning management system.*
    - *IT Manager*
      - *Needs: Secure, scalable hosting; smooth integration with existing databases.*
      - *Constraints: Must use university-approved tech stack.*
      - *Expectations: High uptime, easy maintenance.*
    - *Finance Officer*
      - *Needs: Automated tuition calculation linked to enrolment.*
      - *Constraints: Compliance with finance regulations.*
      - *Expectations: Real-time financial reporting.*
    - *University Administration*
      - *Needs: Reporting dashboards on enrolment trends.*
      - *Constraints: Budget and project deadlines.*
      - *Expectations: Demonstrable ROI and improved student satisfaction.*
  - Stakeholders need to provide needs, constraints, and expectations.
  - PM is to compile top 5 functional and non-functional requirements alongside the stakeholders.

Part C – Lecture Recap

List out and discuss the following:

- Requirement gathering strategies Prototyping; User Story Mapping
- Stakeholder analysis strategies Power-Interest Grid